

When Correlation Means Causation: Pragmatic Factors Modulate Causal Implicatures in Decision-Making Contexts

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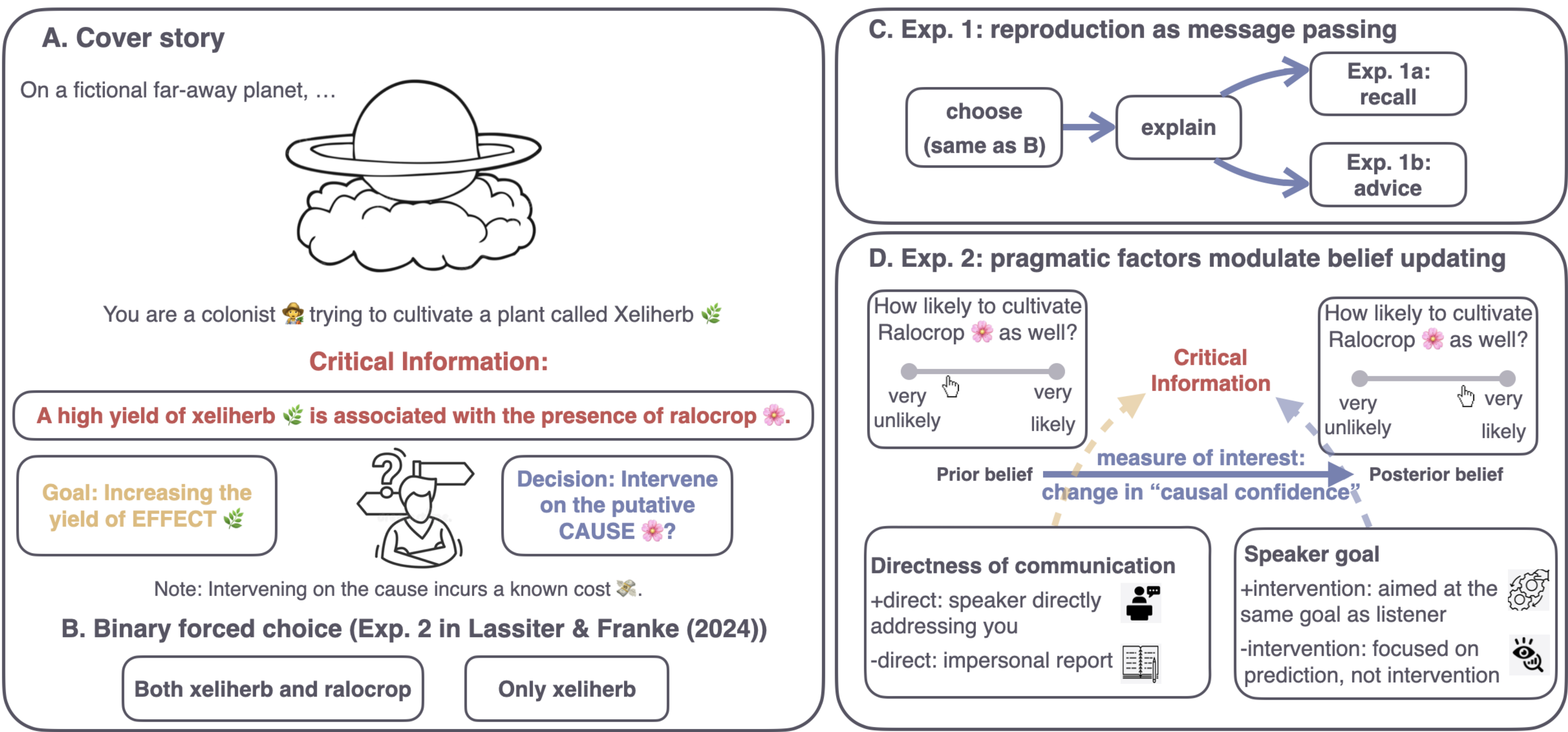
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Paradigm



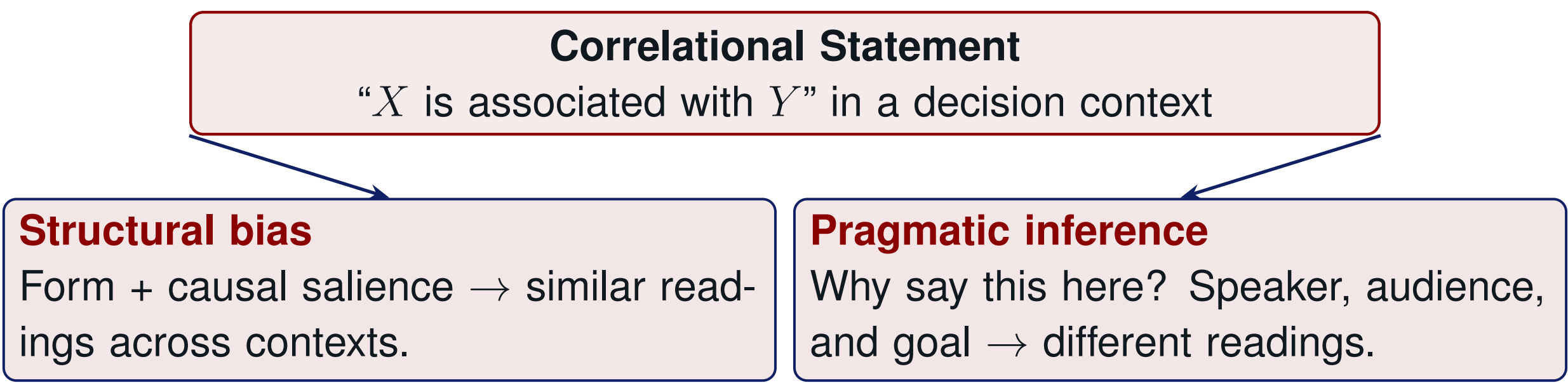
Action proxy: choosing both plants means acting on the putative cause.

Language proxy: causal paraphrases show how participants interpret the message.

Belief proxy: prior-posterior slider change tracks causal belief updating.

Motivation and Accounts

- Correlational statements can evoke causal interpretations and guide decisions, even without an explicit causal claim.
- We ask whether this enrichment reflects stable linguistic form or pragmatic reasoning about why the speaker used this message here.



Experiment 1: Message Passing

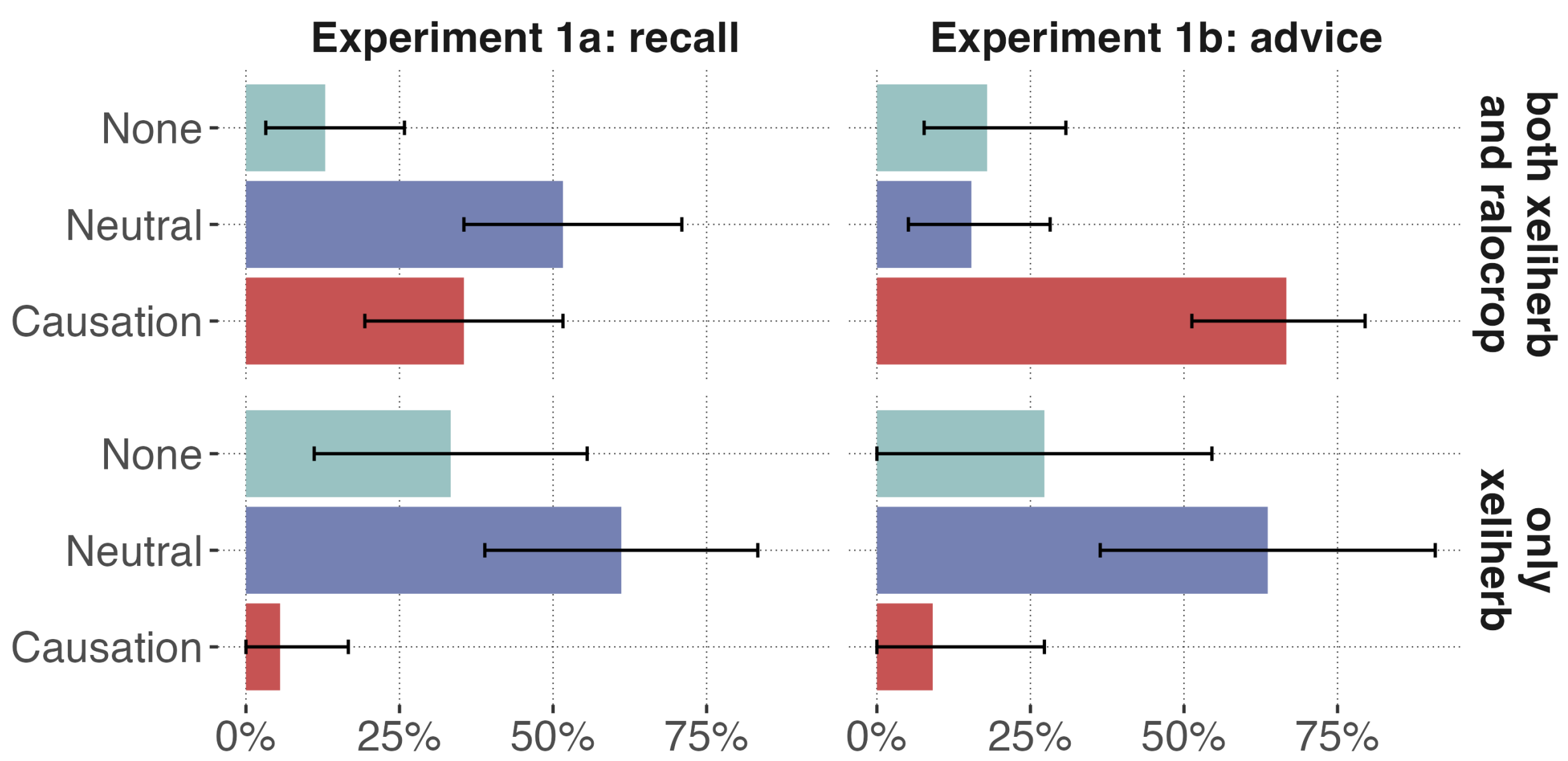
Sample: $N = 99$

Task: 1a: recall report; 1b: advise outpost

Coding: responses coded *causal* / *neutral* / *none*

Result: Participants who chose both plants produced more causal paraphrases, especially when advising another outpost.

Model check: the both-vs-only contrast was credibly positive in both recall and advice.



References and Acknowledgments

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Experiment 2: Belief Updating

Sample

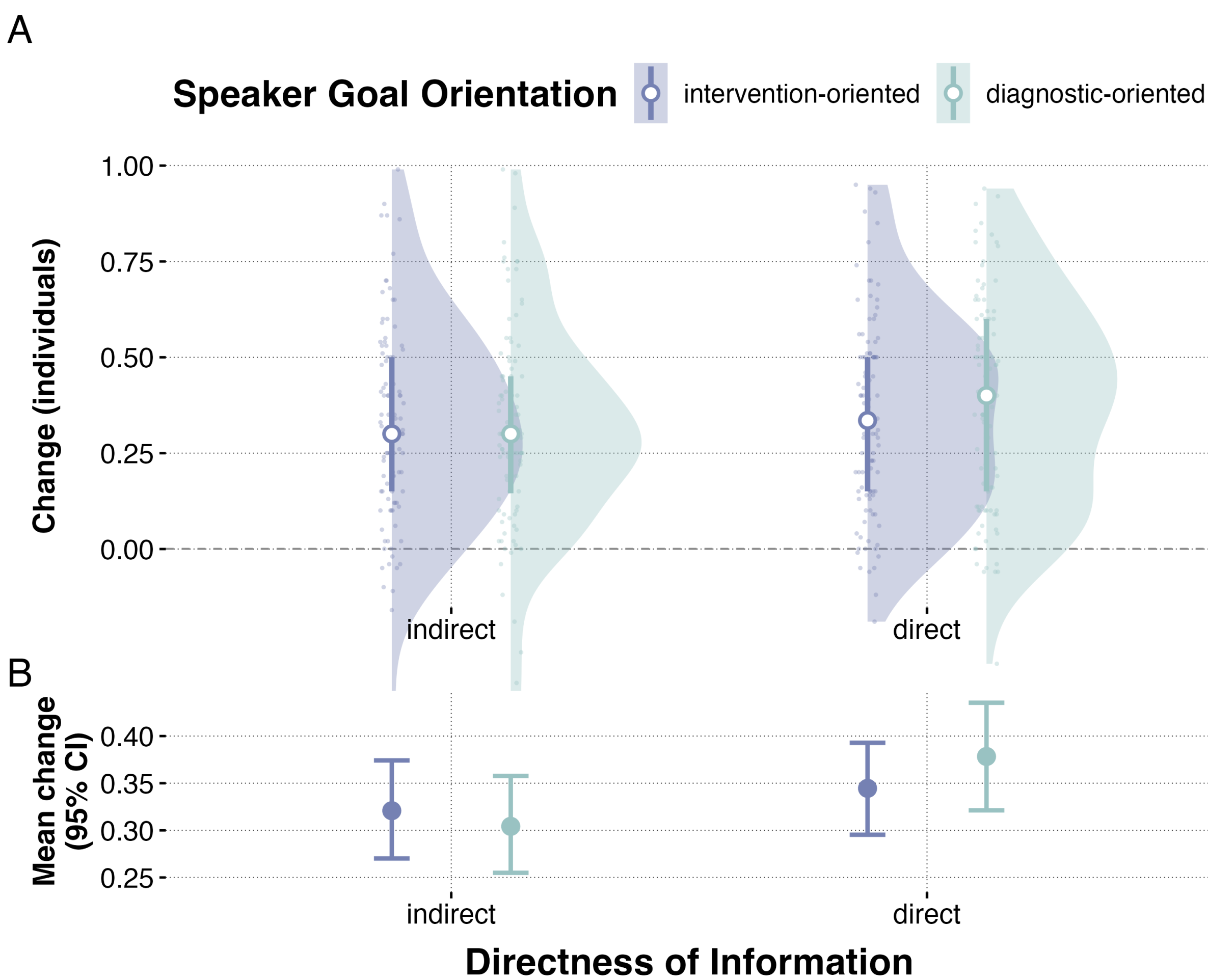
$N = 404$

Procedure

prior rating → report → posterior rating

Measure

change: posterior – prior



Design and pragmatic cues

Directness: addressed report [evidence selected for this decision] vs. impersonal old note [archival evidence; weak audience design].

Speaker goal: cultivation team [intervention-oriented; improve farming] vs. localization team [diagnostic-oriented; locate *xeliherb*].

Result: The correlational report generally increased willingness to intervene on the putative cause. Pragmatic context modulated the size of this update: in direct reports, belief increased more when the speaker's goal was not intervention-aligned, consistent with pragmatic reasoning about why this speaker chose this message for this audience; in indirect reports, the goal difference was small and tended toward the explicitly intervention-aligned farming context.

Model check: overall posterior-minus-prior change was $+0.31$ [0.28, 0.34] on the 0–1 rating scale. Direct reports added $+0.06$ [0.02, 0.09], and the directness-by-goal interaction was credible ($+0.07$, $P = .97$).

Conclusion

- Correlational information shifted causal belief:** willingness to intervene increased after the correlational message.
- Pragmatic context mattered:** audience design and speaker goals modulated causal belief updating.